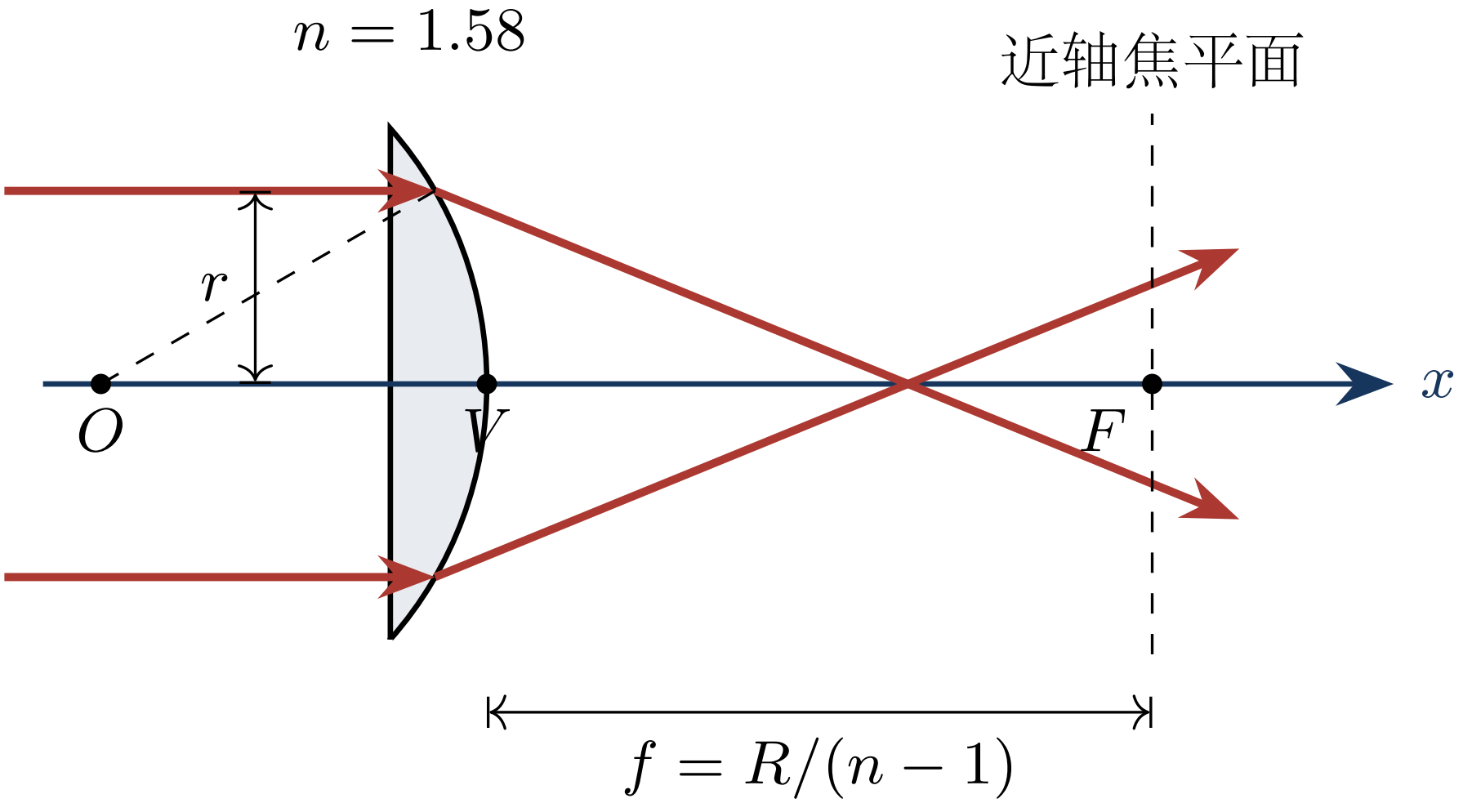
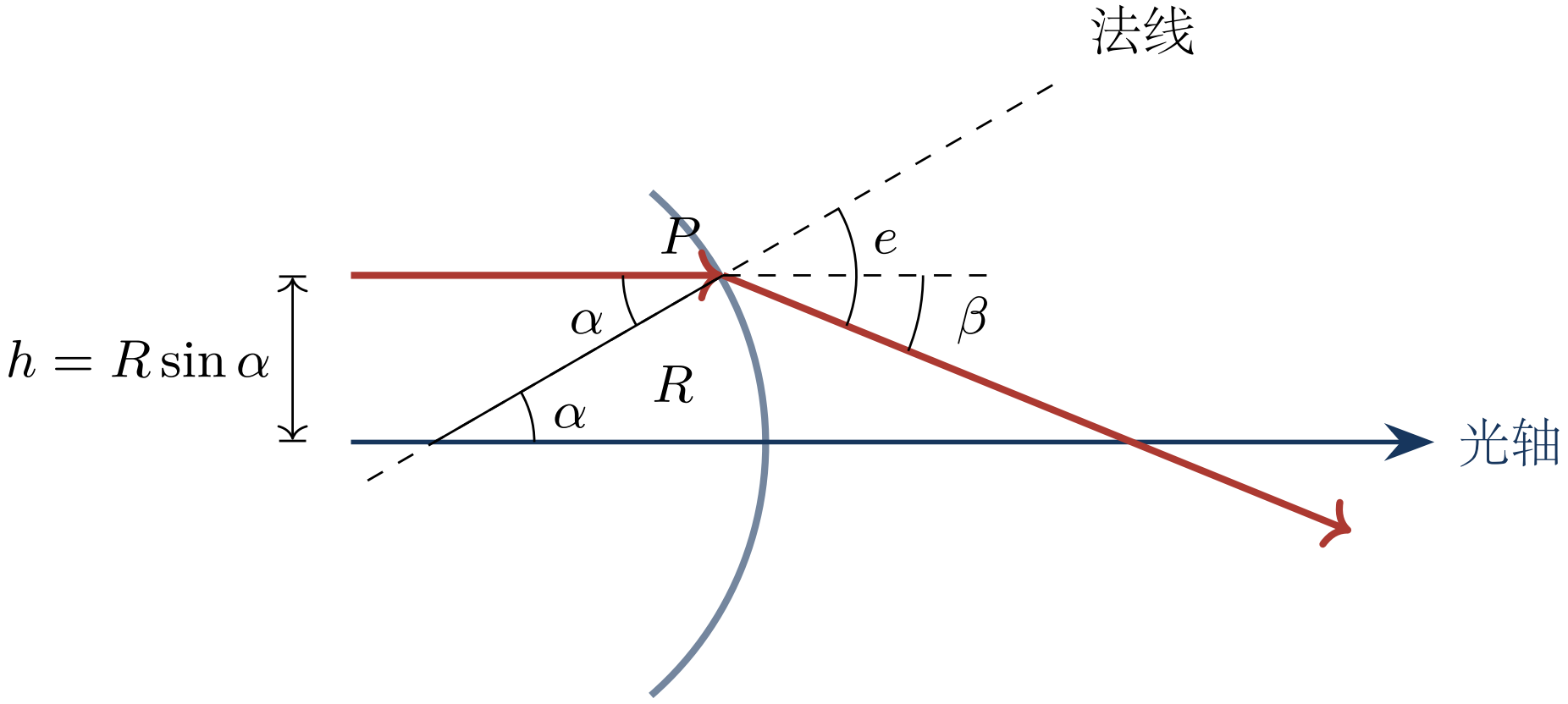
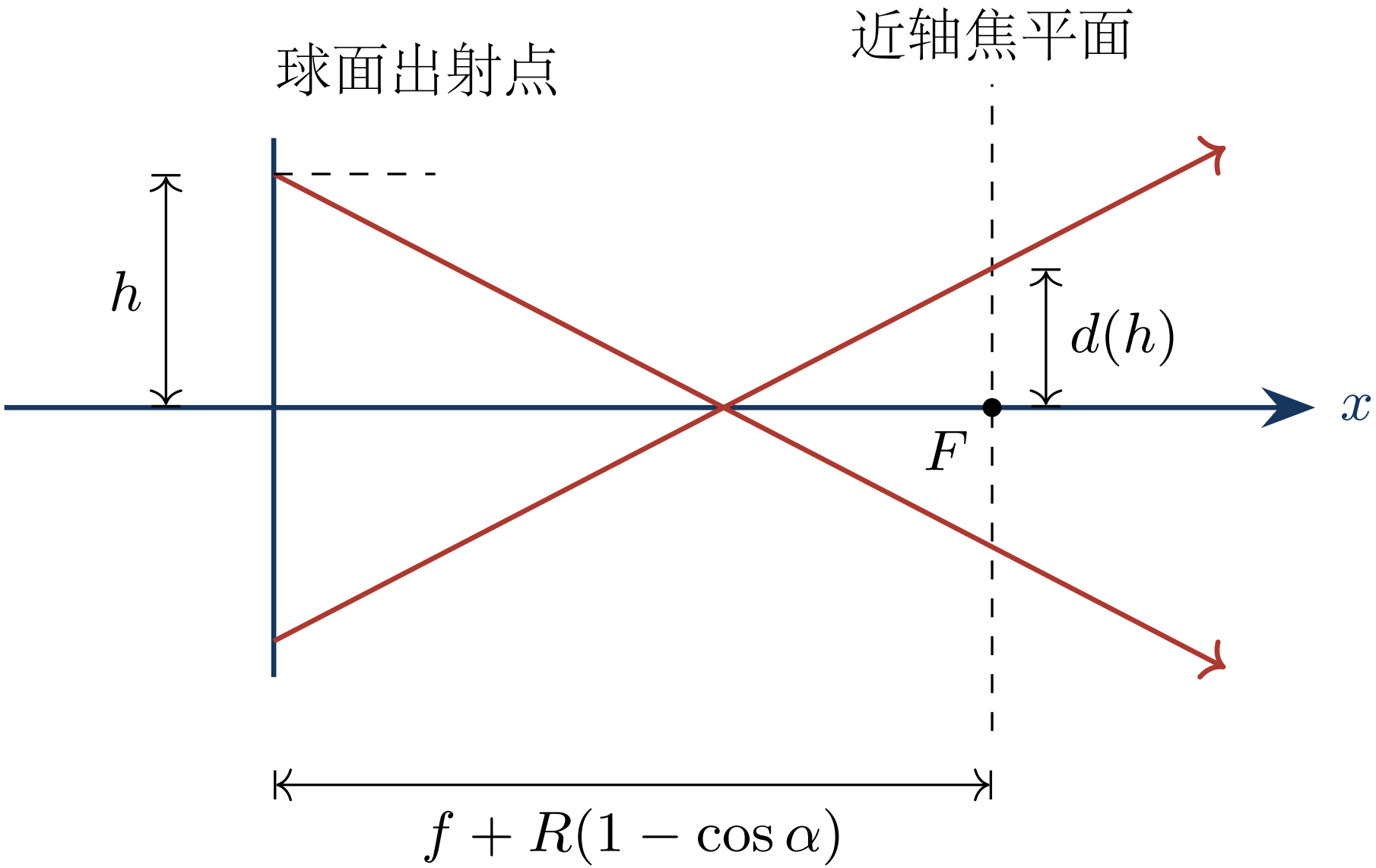
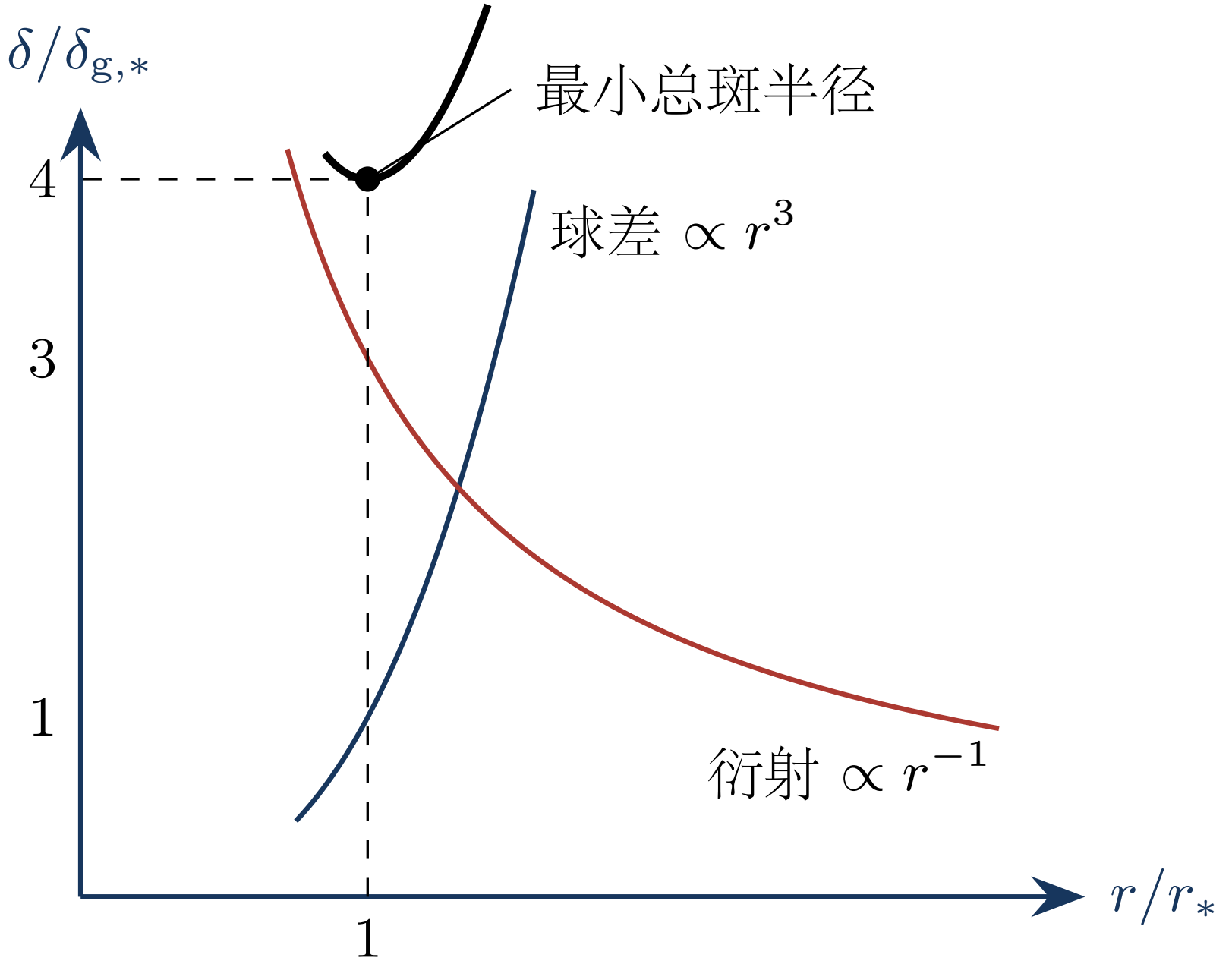


平面先入射，球面后出射；边缘光线提前交轴

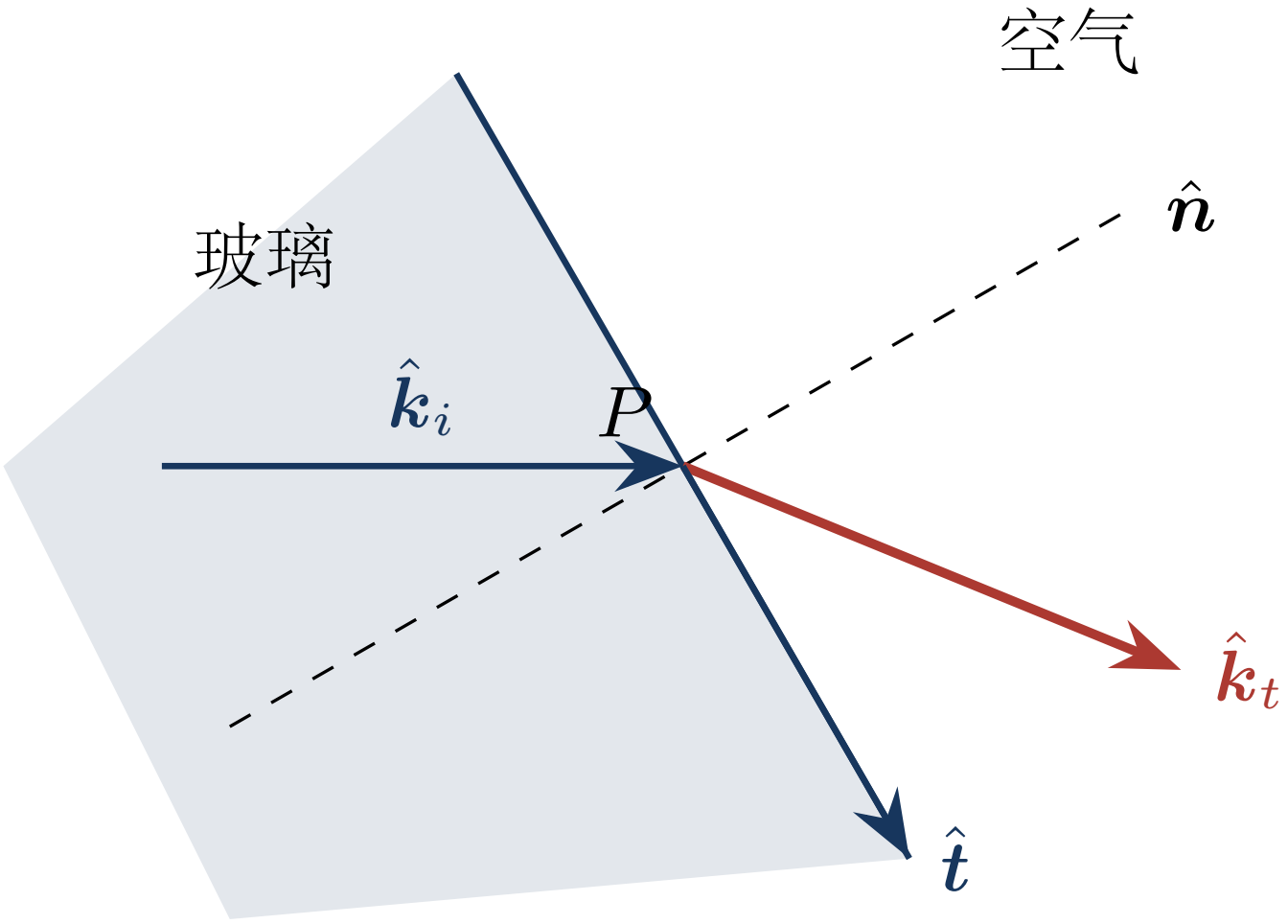

法线
P
e
α
β
h = R sin α
R
α
光轴
sin e = n sin α,
β = e − α



$$d = [f + R(1 - \cos\alpha)] \tan\beta - h$$



最优时 $\delta_{\mathrm{d}} = 3\delta_{\mathrm{g}}$



$$n(\hat{\boldsymbol{k}}_i \cdot \hat{\boldsymbol{t}}) = \hat{\boldsymbol{k}}_t \cdot \hat{\boldsymbol{t}}$$

法向透射分量取向空气的一支